

Hannes Stärk

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Education

Massachusetts Institute of Technology (MIT) , PhD in EECS Machine Learning	June 2022 – present
Advised by Prof. Tommi Jaakkola and Prof. Regina Barzilay.	
Technical University of Munich (TUM) , M.Sc. Informatics Machine Learning	Oct 2019 – Sept 2021
Passed with high distinction (1.2); no corrections for Master's thesis.	
Bundeswehr University Munich , B.Sc. Informatics Mathematics Track	Oct 2017 – Sept 2019
Only student to complete the 3-year curriculum in 2 years.	

Teaching Experience

Modeling with Machine Learning: From Algorithms to Applications — MIT	Spring 2025
Instructor. Designed lectures, problem sets, and exams; held office hours.	
Operations Research — Technical University of Munich, Decision Sciences	Apr 2021 – Sept 2021
Instructor. Taught two recitations per week for ~40 students; supported online forum.	
Deep Learning — Technical University of Munich, CV & AI Niessner Lab	Nov 2020 – Apr 2021
Instructor. Weekly office hours; created exercises and Jupyter notebooks; answered questions in online forum.	

Work Experience

Boltz PBC — Founding member of technical staff	April 2025 – present
Valence Discovery — ML Research Intern	Mar 2022 – May 2022
BIB Augsburg gGmbH — Mathematics Instructor (Augsburg, DE)	Feb 2020 – Nov 2021
4h/week: teaching linear algebra, analysis, and statistics; online lectures and weekly individual lessons.	
Bundeswehr University Munich — Student Assistant (Munich, DE)	Sept 2018 – July 2019
10h/week: causal inference and structure learning in Bayesian networks.	
Allianz Deutschland AG — Dual Study Program (Munich, DE)	Sept 2017 – Sept 2019
38h/week: web development and digital infrastructure in an agile team; technical training in computer science.	

Research Dissemination

Starkly Speaking Reading Group — Organizer (virtual)	Aug 2021 – present
Weekly Zoom discussions with paper authors; ~75 attendees; Slack community >4,000 researchers.	
Molecular Machine Learning Conference (MIT) — Founder (Cambridge, MA)	Oct 2022 – present
~300-attendee conference in honor of Octavian Ganea.	
Learning on Graphs Conference — Co-founder (virtual)	Dec 2022 – Dec 2024
Reviewing with financial incentives; decentralized meetups; free registration; fully open access.	

Invited Talks

Stanford AI+Biomedicine Seminar	Nov 2025
University of California, San Francisco	Nov 2025
Stanford AI+Biomedicine Seminar	Jul 2024
American Chemical Society (Skolnik Award Symposium)	Aug 2023

Publications

Note: * denotes equal first authorship; † denotes co-corresponding authorship.

Highlighted Preprints

BoltzGen: Toward Universal Binder Design 2025
H. Stärk[†], F. Faltings, M. G. Choi, Y. Xie, E. Hur, T. O'Donnell, A. Bushuiev, T. Uçar, S. Passaro, W. Mao, M. Reveiz, R. Bushuiev, T. Pluskal, J. Sivic, K. Kreis, A. Vahdat, S. Ray, J. T. Goldstein, A. Savinov, J. A. Hambalek, A. Gupta, D. A. Taquiri-Diaz, Y. Zhang, A. K. Hatstat, A. Arada, N. H. Kim, E. Tackie-Yarboi, D. Boselli, L. Schnaider, C. C. Liu, G.-W. Li, D. Hnisz, D. M. Sabatini, W. F. DeGrado, J. Wohlgend, G. Corso, R. Barzilay, T. Jaakkola. *Preprint*.

Boltz-2: Towards accurate and efficient binding affinity prediction 2025
S. Passaro*, G. Corso*, J. Wohlgend*, M. Reveiz*, S. Thaler*, V. R. Somnath, N. Getz, T. Portnoi, J. Roy, H. Stärk, D. Kwabi-Addo, D. Beaini, T. Jaakkola, R. Barzilay. *Preprint*.

Conference Papers (NeurIPS / ICML / ICLR)

ProtComposer: Compositional Protein Structure Generation with 3D Ellipsoids 2025
H. Stärk*, B. Jing*, T. Geffner, J. Yim, T. Jaakkola, A. Vahdat, K. Kreis. *ICLR Oral. Among 0.7% best scores*.

Think while you generate: Discrete diffusion with planned denoising 2025
S. Liu, J. Nam, A. Campbell, H. Stärk, Y. Xu, T. Jaakkola, R. Gómez-Bombarelli. *ICLR*.

ETFlow: Equivariant Flow-Matching for Molecular Conformer Generation 2025
M. Hassan, N. Shenoy, J. Lee, H. Stärk, S. Thaler, D. Beaini. *NeurIPS*.

Dirichlet Flow Matching with Applications to DNA Sequence Design 2024
H. Stärk*, B. Jing*, C. Wang, G. Corso, B. Berger, R. Barzilay, T. Jaakkola. *ICML*.

Harmonic Self-Conditioned Flow Matching for Multi-Ligand Docking and Binding Site Design 2024
H. Stärk*, B. Jing*, R. Barzilay, T. Jaakkola. *ICML*.

Generative Modeling of Molecular Dynamics Trajectories 2024
B. Jing*, H. Stärk*, T. Jaakkola, B. Berger. *NeurIPS*.

DiffDock: Diffusion Steps, Twists, and Turns for Molecular Docking 2023
G. Corso*, H. Stärk*, B. Jing*, R. Barzilay, T. Jaakkola. *ICLR*.

EquiBind: Geometric Deep Learning for Drug Binding Structure Prediction 2022
H. Stärk*, O. E. Ganea*, L. Pattanaik, R. Barzilay, T. Jaakkola. *ICML*.

3D Infomax improves GNNs for Molecular Property Prediction 2022
H. Stärk, D. Beaini, G. Corso, P. Tossou, C. Dallago, S. Günnemann, P. Liò. *ICML*.

Journal Articles

Evolutionary design of conductive pathways using a generative autoencoder 2025
M. C. Ignuta-Ciuncanu, H. Stärk, R. F. Martinez-Botas. *International Communications in Heat and Mass Transfer* 166:109098.

Graph neural networks 2024
G. Corso*, H. Stärk*, S. Jegelka, T. Jaakkola, R. Barzilay. *Nature Reviews Methods Primers* 4(1):17.

Diffusion models in protein structure and docking 2024
J. Yim, H. Stärk, G. Corso, B. Jing, R. Barzilay, T. S. Jaakkola. *WIREs Comput. Mol. Science* 14(2):e1711.

- Blind protein-ligand docking with diffusion-based deep generative models** 2023
G. Corso, H. Stärk, R. Barzilay, T. Jaakkola. *Biophysical Journal* 122(3):143a.
- Benchmarking AlphaFold-enabled molecular docking predictions for antibiotic discovery** 2022
F. Wong, A. Krishnan, E. J. Zheng, H. Stärk, A. L. Manson, A. M. Earl, T. Jaakkola, J. J. Collins. *Molecular Systems Biology* 18(9):e11081.
- Equivariant 3D-Conditional Diffusion Models for Molecular Linker Design** 2022
I. Igashov, H. Stärk, C. Vignac, V. G. Satorras, P. Frossard, M. Welling, M. Bronstein, B. Correia. *Nature Machine Intelligence* 6(4):417–427.
- Light Attention Predicts Protein Location from the Language of Life** 2021
H. Stärk, C. Dallago, M. Heinzinger, B. Rost. *Bioinformatics Advances*.

Preprints

- ProxelGen: Generating Proteins as 3D Densities** 2025
F. Faltings*, H. Stärk*, R. Barzilay, T. Jaakkola. *Preprint*.
- Protein FID: Improved Evaluation of Protein Structure Generative Models** 2025
F. Faltings, H. Stärk, T. Jaakkola, R. Barzilay. *Preprint*.
- Training on test proteins improves fitness, structure, and function prediction** 2024
A. Bushuiev, R. Bushuiev, N. Zadorozhny, R. Samusevich, H. Stärk, J. Sedlar, T. Pluskal, J. Sivic. *Preprint*.
- CodonMPNN for Organism Specific and Codon Optimal Inverse Folding** 2024
H. Stark, U. Padia, J. Balla, C. Diao, G. Church. *Preprint*.
- PINDER: The protein interaction dataset and evaluation resource** 2024
D. Kovtun, M. Akdel, A. Goncarenko, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, E. Rossi, G. Corso, H. Stärk, Z. Carpenter, E. Kucukbenli, M. Bronstein, L. Naef. *Preprint*.
- PLINDER: The protein-ligand interactions dataset and evaluation resource** 2024
J. Durairaj, Y. Adeshina, Z. Cao, X. Zhang, V. Oleinikovas, T. Duignan, Z. McClure, X. Robin, G. Studer, D. Kovtun, E. Rossi, G. Zhou, S. Veccham, C. Isert, Y. Peng, P. Sundareson, M. Akdel, G. Corso, H. Stärk, G. Tauriello, Z. Carpenter, M. Bronstein, E. Kucukbenli, T. Schwede, L. Naef. *Preprint*.
- FreeFlow: Latent Flow Matching for Free Energy Difference Estimation** 2024
E. Erdogan*, R. Ralev*, M. Rebensburg*, C. Marquet, L. Klein, H. Stark. *Preprint*.
- Transition Path Sampling with Boltzmann Generator-based MCMC Moves** 2023
M. Plainer*, H. Stärk*, C. Bunne, S. Günnemann. *Preprint*.
- DiffDock-Pocket: Diffusion for Pocket-Level Docking with Sidechain Flexibility** 2023
M. Plainer*, M. Toth*, S. Dobers*, H. Stark, G. Corso, C. Marquet, R. Barzilay. *Preprint*.
- DiffDock-PP: Rigid Protein-Protein Docking with Diffusion Models** 2023
M. A. Ketata*, C. Laue*, R. Mammadov*, H. Stärk, M. Wu, G. Corso, C. Marquet, *et al.* *Preprint*.
- Latent Space Simulator for Unveiling Molecular Free Energy Landscapes and Predicting Transition Dynamics** 2023
S. Dobers, H. Stark, X. Fu, D. Beaini, S. Günnemann. *Preprint*.
- Generalized Laplacian Positional Encoding for Graph Representation Learning** 2022
S. Maskey, A. Parviz, M. Thiessen, H. Stärk, Y. Sadikaj, H. Maron. *Preprint*.
- Graph Anisotropic Diffusion** 2022
A. A. Elhag, G. Corso, H. Stärk, M. M. Bronstein. *Preprint*.
- Jointly Learnable Data Augmentations for Self-Supervised GNNs** 2021
Z. T. Kefato, S. Girdzijauskas, H. Stärk. *Preprint*.